

Lecture 9.

- Announcement
- pdf cdf E.V., $h(\mathcal{Z})$

$$\underline{E(a\mathcal{Z} + b) = a E(\mathcal{Z}) + b.}$$

$$\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_n \quad (S, \mathbb{F}, \mathbb{P}).$$

$$\bullet \quad E(\mathcal{Z}_1 + \dots + \mathcal{Z}_n) = E(\mathcal{Z}_1) + \dots + E(\mathcal{Z}_n).$$

holds whether or not $\mathcal{Z}_i$ independent.

$$\bullet \quad A \quad \mathbb{1}_A(w) = \begin{cases} 1 & , w \in A \\ 0 & , w \notin A. \end{cases}$$

$$E(\mathbb{1}_A) = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A).$$

Example. (Runs in a shuffled deck).
26 red 26 black.

RRBRRRB

$$E(\text{runs}) = ?$$

C_1 1.2.

S_2 .

C_{51}

51-th

52-th

$\mathbb{1}_{C_i}$

$C_i = \{ \text{cards } i \text{ and } i+1 \text{ differ in color} \}$
 $i = 1, 2, \dots, 51.$

$$\text{runs} = \underline{1} + \sum_{i=1}^{51} \mathbb{1}_{C_i}$$

$$E(\text{runs}) = 1 + \sum_{i=1}^{51} E(1_{C_i})$$

$$= 1 + \sum_{i=1}^{51} \underline{P(C_i)}$$

$$= 1 + 51 \times \frac{26}{52} \times \frac{26}{51} \times 2$$

$$\underline{P(C_i)} = \frac{26}{52} \cdot \frac{26}{\textcircled{51}} \cdot 2$$

BB
BR

BB

$$\begin{aligned} P(RB) &= \underline{P(R \text{ in the first})} \cdot \underline{P(B \text{ in the second} \mid R \text{ in the first})} \\ &= \frac{26}{52} \cdot \end{aligned}$$

(Tail-Sum formula)

ξ takes values in $\{0, 1, 2, \dots\}$.

then $\boxed{E(\xi) = \sum_{k=1}^{\infty} P\{\xi \geq k\}}$

Proof

$$E(\xi) = \sum_{n=0}^{\infty} n \cdot \underline{P\{\xi = n\}}$$

$$= \sum_{\textcircled{n=0}}^{\infty} \sum_{\textcircled{k=1}}^n P\{\xi = n\}$$

$$= \sum_{\textcircled{k=1}}^{\infty} \sum_{\textcircled{n=k}}^{\infty} P\{\xi = n\}$$

$$= \sum_{k=1}^{\infty} P\{\xi \geq k\}$$

$$\textcircled{1 \leq k \leq n} \\ 0 \leq n < \infty$$

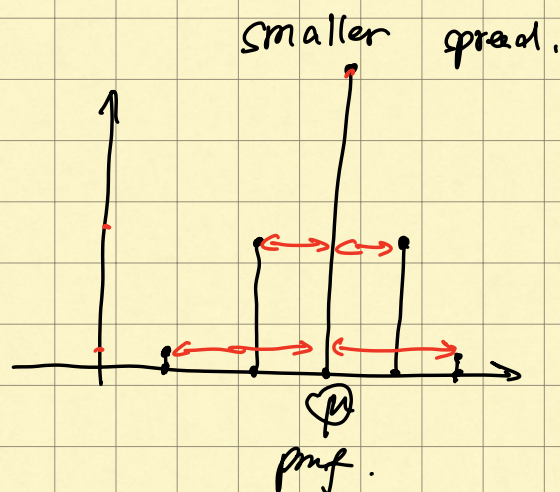
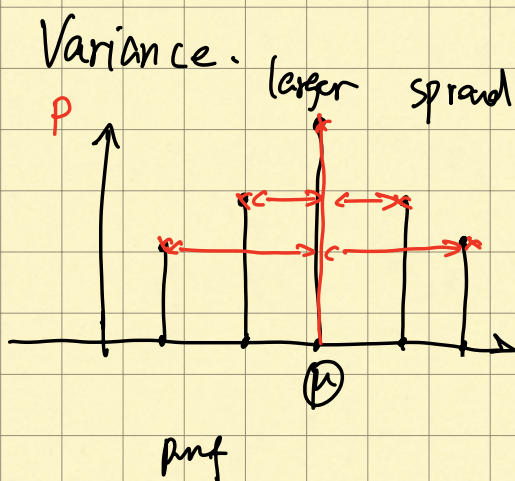
ξ = number of shots till I hit the target.

$$\mathbb{P}\{\xi = k\} = (1-\beta)^{k-1} \beta$$

$\{\xi \geq k\}$ means the first $k-1$ shots missed.

$$\mathbb{P}\{\xi \geq k\} = (1-\beta)^{k-1}$$

$$\mathbb{E}(\xi) = \sum_{k=1}^{\infty} \mathbb{P}\{\xi \geq k\} = \sum_{k=1}^{\infty} (1-\beta)^{k-1} = \frac{1}{1-(1-\beta)} = \frac{1}{\beta}.$$



Def (Variance) ξ $\mathbb{E}(\xi) = \mu$.

The variance of ξ

$$\mathbb{E}(|\xi - \mu|).$$

$$V(\xi) = \text{Var}(\xi) = \sigma_{\xi}^2 := \mathbb{E}((\xi - \mu)^2).$$

σ_{ξ} is the standard deviation.

$$\sigma_{\xi} = \sqrt{\text{Var}(\xi)}.$$

$$\text{Var}(\xi) = \mathbb{E}((\xi - \mu)^2)$$

$$= E(\xi^2) - (E(\xi))^2$$

Proof: $E((\xi - \mu)^2) = E(\xi^2 - 2\mu\xi + \mu^2)$
 $= E(\xi^2) - 2\mu E(\xi) + \mu^2$
 $= E(\xi^2) - 2\mu^2 + \mu^2$
 $= E(\xi^2) - \mu^2$
 $= E(\xi^2) - E(\xi)^2$

$$E(\xi), E(\xi^2)$$

$$\text{Var}(a\xi + b) = a^2 \text{Var}(\xi)$$

binomial distribution.
 geometric
 hypergeometric
 Negative binomial
 poisson

discrete.

Experiment
 Experiment 1, Experiment 2.

(Independent Trials). I conduct an experiment.
 consisting of independent smaller experiments,
 each smaller experiment is called an independent trial.

Binomial Distribution.

An experiment consists of n independent trials.

- each trial has exactly two outcomes:

Success (S) or Failure (F).

- $P(S) = p$, $P(F) = 1 - p \stackrel{\text{def}}{=} q$, the same on every trial.

$$S := \{ HHTHT \dots T, \dots, HTHT \dots T \}$$

$\xi :=$ the number of S among the n trials.

Then ξ is called Binomial distribution.

$$\xi \sim \text{Binomial}(n, p)$$

pmf of ξ is written as $b(x; n, p)$.

$$b(x; n, p) = P\{\xi = x\}$$

$$= \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x=0, \dots, n \\ 0, & \text{else} \end{cases}$$

$$\underbrace{S S F F S \dots S}_{x \text{ S's}} \binom{n}{x}$$

$$E(\xi) = np$$

$$\text{Var}(\xi) = np(1-p).$$

(Geometric distribution).

In binomial experiment, $\xi =$ the number of trials till the first S.

$\xi \sim \text{Geometric}(p).$

$$\mathbb{P}\{\xi = x\} = g(x; p) = (1-p)^{x-1} p$$
$$x = 1, 2, 3, \dots$$

$$E(\xi) = \frac{1}{p}, \quad \text{Var}(\xi) = \frac{1-p}{p^2}.$$